

CHAINSILIENCE AI: An Agentic, Digital-Twin Framework for Supply-Chain Disruption Intelligence

Ka Woon Ho, Ke Xin Cao, Gary Leng, Mo Xia, Andrew Li

ansonho31@gmail.com
caokexin5632@gmail.com
Lengg0701@gmail.com
mark.xia.mo@gmail.com
andrew.li3056@gmail.com

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AI-powered resilience for SMEs, startups, and semiconductor supply networks

Abstract — Chainsilience AI is a proposed disruption-intelligence and response platform for small and medium-sized enterprises (SMEs), startups, and semiconductor-related businesses. It converts fragmented economic, geopolitical, environmental, regulatory, logistics, and industry reporting into company-specific operational decisions. The pipeline collects external information, extracts and verifies events with AI agents, maps them to a supply-chain knowledge graph and digital twin, estimates exposure and operational impact, and produces explainable recommendations, draft communications, dashboard updates, and alerts. Unlike generic news monitoring, the system combines public evidence with private enterprise data such as suppliers, sites, routes, inventory, production schedules, and customer orders. This paper defines the architecture, risk model, governance controls, evaluation protocol, and a retrospective Taiwan-earthquake scenario. Chainsilience AI is currently implemented as a functional prototype demonstrating news relevance filtering, disruption extraction, company-specific exposure analysis, mitigation recommendation, and dashboard presentation. Advanced enterprise integrations, large-scale evaluation, and controlled external actions remain planned capabilities.

Keywords- supply-chain resilience, agentic AI, digital twin, knowledge graph, semiconductor risk, retrieval-augmented generation, LLM reasoning, SME, model context protocols, natural language processing

I. INTRODUCTION AND PROBLEM

Modern supply chains are exposed to interacting shocks rather than isolated failures. In 2024, disruption around the Red Sea, Suez Canal, and Panama Canal lengthened routes, increased freight-rate volatility, and raised costs [1]. Semiconductor ecosystems add concentrated materials, specialized facilities, long qualification cycles, and geopolitical controls; a U.S. industrial-base assessment identifies fragile supply chains, customer concentration, geopolitical factors, and limited transparency as material risks [2].

Large enterprises may purchase specialist intelligence and maintain dedicated risk teams. Many SMEs instead depend on supplier emails, spreadsheets, search alerts, and periodic meetings. The decisive information may exist, but it is separated from the operational context required to answer: Which order is threatened? How long will stock last? Which customer commitment fails first? What is the least-cost mitigation strategy?

A. Problem Statement

Manual monitoring creates three delays. First, analysts must discover a relevant event among high-volume reporting. Second, they must verify location, affected entities, products, timing, and severity. Third, they must translate the event into dependencies hidden across business

systems. Conventional disruption models support preparedness and recovery, but their value depends on timely, current inputs [3]. Chainsilience AI addresses this event-to-decision gap.

B. Research Gap

Digital supply-chain twins can represent the network state and test disruption scenarios [4], while ISO 23247 provides general digital-twin principles for manufacturing [5]. Knowledge graphs can reveal multi-tier dependencies between entities such as suppliers and shipping routes and support risk reasoning [6], and recent work shows that large language models (LLMs) can extract supply-chain related news from public sources [7]. The remaining design challenge is to synthesize these capabilities into a governed operational workflow that preserves evidence, quantifies uncertainty, and requires authorization before consequential actions.

C. Objectives

The project has five objectives: (1) detect and normalize relevant external events; (2) connect events to enterprise assets and commitments; (3) estimate revenue at risk, delay, inventory, cost of mitigation, and production stoppage probability; (4) generate traceable mitigation options and communications; and (5) evaluate timeliness, accuracy, usefulness, and safety.

D. Scope and Status

Implemented/prototyped: intended for demonstrated modules and interfaces only after they exist. **Planned:** architecture, integrations, scoring, simulations, and controlled agent actions described here. **Assumed:** enterprise data availability, access rights, and scenario inputs. **Illustrative:** all case-study probabilities, costs, and recommended allocations. This explicit labeling prevents a design proposal from being mistaken for production evidence.

Question	System output
What happened?	Verified event + source provenance
Why relevant?	Mapped suppliers, parts, sites, routes
What follows?	Delay, days of cover, order impact
What now?	Ranked actions, cost, trade-offs, owner

TABLE I. FROM EXTERNAL SIGNAL TO OPERATIONAL DECISION

II. CHAINSILIENCE AI SOLUTION AND CONTRIBUTION

A. Value Proposition

Chainsilience AI is a continuously refreshed decision-support layer above a firm's existing systems. It does not replace procurement, enterprise resource planning (ERP), manufacturing execution, transport management, or human judgment. It combines public evidence with authorized internal data and presents the smallest set of decisions needed to protect service, margin, and continuity.

The primary users are procurement managers, operations planners, startup founders, logistics teams, finance controllers, and risk owners. A user can move from an alert to the supporting articles, extracted facts, affected graph path, assumptions, simulation result, recommended action, and approval history. This evidence chain is essential because the same event may be severe for one company and irrelevant for another, depending on their respective suppliers, routes, ports, etc.

B. Differentiators

Capability	Generic monitor	Chainsilience AI
Signal	Keyword/news alert	LLM relevance filtering and cross-source verification
Context	Sector or geography	Company graph + digital twin
Impact	Narrative severity	Orders, delay, revenue at risk, inventory
Response	Analyst interpretation	Simulation of multi-versioned response options
Audit	Link or summary	Claim-level provenance + approvals

TABLE II. DIFFERENTIATION FROM GENERIC MONITORING

The contribution is not a claim that an LLM can autonomously manage a supply chain. It is a governed composition of retrieval, structured extraction, deterministic calculations, graph traversal, simulation, optimization, and human authorization. LLMs handle language-intensive tasks; calculators and solvers handle quantities; policy engines determine which actions are permitted.

C. Stakeholder Outcomes

For SMEs, the intended outcome is earlier, more consistent analysis without a large intelligence team. For semiconductor firms, the graph can represent foundries, outsourced assembly and test, materials, equipment, utilities, routes, and qualification constraints. For customers and suppliers, templated communications reduce response time while preserving approval.

D. End-to-end Workflow

Execution begins when an authorized company administrator defines the permitted data boundary and uploads or connects the company's product records, bills of materials, component mappings, suppliers and manufacturing sites, approved alternatives, inventory and in-transit stock, purchase and customer orders, production schedules, logistics routes, lead times, costs, and service priorities.

The platform validates and standardizes these records, filters out contradicting or faulty information after review, and creates a versioned supply-chain knowledge graph and digital-twin baseline. Chainsilience AI then monitors authorized external sources, verifies potential disruptions, identifies affected dependency paths, and simulates the operational and financial consequences of taking no action.

The LLM proposes evidence-backed candidate mitigation actions, such as reallocating inventory, rerouting, adjusting production schedules, or transferring orders to qualified alternative suppliers. Deterministic and optimization services compare the candidates using impact metrics such as reduced risk, recovery time, and implementation cost.

A designated decision-maker reviews the supporting sources, affected supply-chain relationships, assumptions, calculations, and alternative strategies. The decision-maker may revise assumptions, modify the proposed response, rerun the simulation, and approve the exact mitigation strategy and any related communications.

Once approved, Chainsilience AI initiates only the actions permitted by company policy, such as creating procurement tasks, preparing supplier emails, notifying internal teams, or passing authorized changes to connected enterprise systems. The dashboard then displays the original no-action risk, projected post-mitigation risk, implementation progress, and realized operational impact as supplier confirmations, inventory updates, shipment information, and production data become available.

Projected risk reduction is clearly distinguished from realized risk reduction until supporting operational evidence has been received.

III. SYSTEM ARCHITECTURE AND DATA PIPELINE

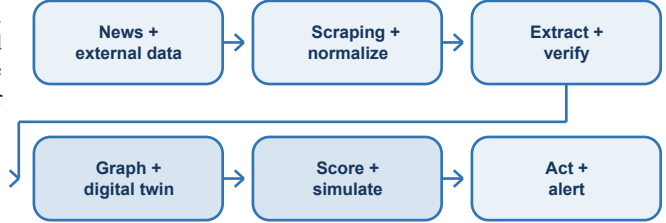


Fig. 1. Core Chainsilience AI pipeline. Each transition stores confidence, provenance, timestamp, and version.

A. News Collection/Scraping and Normalization

Connectors ingest permitted sources: government advisories, earthquake and weather feeds, trade and customs notices, port and carrier updates, company disclosures, trusted media, and licensed datasets. Documents are hashed, timestamped, language-detected, deduplicated, and retained with source metadata. Internal connectors read only the fields required from supplier master data, bills of material, inventory, orders, routes, and schedules.

B. AI-Agent Extraction and Verification

An LLM based extraction AI agent determines whether each collected news item is relevant to the company's suppliers, facilities, routes, products, or components. It converts unstructured text into a schema containing event type, entities, locations, products, start time, expected duration, severity, quantities, and cited evidence spans. A verifier agent filters out potentially unreliable and unsupported information after further review by comparing independent sources, detecting contradictions, checking temporal and geographic consistency, and assigning confidence levels. Retrieval-augmented generation (RAG) grounds summaries in retrieved records [9]; provenance follows the entity-activity-agent concepts standardized by W3C's PROV-O [10].

Agentic workflows may interleave reasoning with tool use, following patterns demonstrated by ReAct [8]. Chainsilience AI constrains this behavior: tools expose typed inputs, least-privilege permissions, timeouts, and auditable results. The LLM cannot directly alter enterprise records or send external communications unless a workflow policy and designated human approver permit it.

C. Enterprise Context Layer

The event object is joined to enterprise identifiers. Entity resolution uses names, addresses, geospatial distance, corporate ownership, product codes, and user-confirmed aliases. Unresolved matches enter a review queue rather than silently becoming graph facts. Confidence therefore travels with each edge and each downstream conclusion.

Layer	Representative data	Primary control
External	Articles, advisories, feeds	Source allowlist + license
Enterprise	BOM, inventory, orders, routes	Role/field access
Reasoning	Events, graph, scenarios	Evidence + confidence
Action	Alerts, drafts, tasks	Approval + audit

TABLE III. DATA LAYERS AND CONTROLS

D. Supply-Chain Knowledge Graph

The graph nodes represent Supplier, Site, Part, Material, Product, Route, Port, Inventory, Production Line, Customer order, Event, and Source. Edges encode relationships such as SUPPLIES, PRODUCED_AT, REQUIRES, SHIPS_VIA, STORED_AT, FULFILLS, LOCATED_IN, OWNED_BY, and AFFECTS. For example, if two nodes are product and route, then the relationship is SHIPS_VIA. Temporal validity permits the system to distinguish a current supplier from an expired contract. Evidence and confidence are attached to every externally inferred relationship.

Graph traversal identifies exposure paths that a flat alert cannot: *earthquake* → *industrial park* → *wafer fab* → *chip family* → *assembly site* → *finished product* → *customer order*. Knowledge-graph reasoning has been studied for discovering underlying supply-chain risks [6], while LLM-based extraction can extend mapping from public information [7]. Chainsilience AI combines those methods with private, authoritative enterprise edges.

E. Digital Twin

The knowledge graph first establishes the enterprise entities and their dependencies. Each operational entity in the graph such as a supplier, inventory, or route, is linked to a corresponding digital twin containing its current and historical operational state. Together, these connected entity twins form the enterprise supply-chain digital twin. It is a time-indexed computational representation of the operational network, not just a dashboard. It combines topology with state variables: on-hand and in-transit inventory, demand, yields, capacities, lead times, route schedules, recovery assumptions, and contractual priorities. Digital supply-chain twins support disruption experiments for risk analysis and mitigation simulations [4].

Entity/state	Example fields
Supplier-site	capacity, recovery curve, qualification
Part/material	BOM usage, substitutes, yield loss
Inventory	on-hand, committed, in-transit, expiry
Route	mode, lane, transit, cost, chokepoints
Order	quantity, due date, priority, penalty

TABLE IV. MINIMUM DIGITAL-TWIN STATE

Twin updates are event-sourced: each change records its source, effective time, ingestion time, and previous value. Scenario runs operate on versioned snapshots so that analysts can reproduce why a recommendation was made.

F. Simulation and Optimization

A discrete-time simulator propagates capacity loss, lead-time shifts, yield changes, transit delays, and allocation rules through the network. Candidate mitigations may include switching suppliers, rerouting, reallocating scarce inventory, rescheduling production, qualifying a substitute, splitting volume, or negotiating due dates. Optimization is constrained by approved suppliers, technical compatibility, minimum order quantities, capacity, cost ceilings, and customer priority.

G. Agent Orchestration

Specialized AI agents have narrow roles: **Scout** (Non-AI) retrieves potential events; **Extractor** structures claims; **Verifier** filters out faulty information; **Mapper** resolves graph entities; **Analyst** requests deterministic scenarios; **Planner** ranks feasible responses; and **Communicator** drafts internal alerts or supplier/customer emails. A policy gateway mediates all tools.

Model Context Protocol (MCP) is one possible interoperability layer for connecting AI applications to external resources, prompts, and tools [17]. In this design, an MCP server may expose a read-only inventory query or a draft-email operation. MCP is not itself a security boundary; authentication, authorization, validation, and approval remain application responsibilities.

Agent	Permitted output	Prohibited by default
Scout	Candidate URLs/events	Bypass source policy
Extractor	Schema + evidence spans	Invent missing fields
Analyst	Scenario request/result	Override calculator
Planner	Candidate mitigation actions	Commit purchase/order
Communicator	Draft message	External send

TABLE V. SEPARATION OF AGENT DUTIES

H. Explainability and Interface

Every alert presents six linked cards: event facts; evidence and source quality; affected graph path; operational assumptions; quantitative impacts; and response options. Confidence is decomposed into source reliability, extraction confidence, entity-match confidence, and model uncertainty. The interface avoids a single opaque 'AI confidence' number.

Each LLM recommended candidate action includes the expected service benefit, implementation cost, recovery time, net financial impacts, and reduced risk. Users may change assumptions and rerun the scenario. A comparison view shows the baseline and each mitigation strategy under the same snapshot.

I. Alert and Communication Flow

Alerts are severity- and role-aware. Low-confidence events request verification, material events notify the designated incident channel, and external communications remain strictly reviewable drafts until approval. The system leverages Natural Language Processing (NLP) to dynamically generate context-aware email drafts that populate key operational details for external stakeholders. For suppliers, NLP constructs targeted inquiries requesting facility status, available capacity, revised lead times, and the next update window. For customers, NLP generates professional status updates that disclose only authorized facts, tailor information to the specific order context, and strictly avoid unsupported certainty.

The design converts AI-generated output into a reviewable draft rather than an irreversible action. Each approval records the reviewer, timestamp, message version, recipients, attached evidence, and downstream tool result. Human approval supports accountable oversight and reduces unauthorized action risk but does not by itself guarantee ethical or correct outcomes. NLP and LLM components strictly provide decision support.

IV. RISK ANALYSIS AND RESPONSE GENERATION

A. Risk Decomposition

Risk is modeled as a set of decision variables, not a free-form LLM judgment. For event e and asset, part, or order i , the normalized score is:

$$R_{i,e} = 100 \times C_e \times [w_H H_e + w_X X_{i,e} + w_V V_i + w_T T_{i,e} - w_M M_{i,e}]$$

Here, C is evidence confidence; H is hazard severity and duration; X is exposure through location, route, or dependency; V is vulnerability from concentration, substitutability, and recovery time; T is time criticality from demand and inventory; and M is mitigation strength. Inputs are scaled to $[0,1]$, w refers to respective weights for each risk factor, which are versioned by use case, and the bracketed term is a weighted sum clipped to $[0,1]$. The score ranks attention. The LLM extracts evidence-backed risk metrics and company-specific factors such as expected delay duration, revenue at risk, inventory coverage, and the absence of qualified alternative suppliers, and structures them as inputs for the deterministic risk engine (the equation above). After the score is calculated, the LLM reasons how different factors synthesized to produce the risk score and risk level.

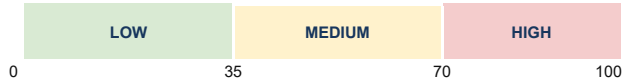


Fig. 2. Configurable triage bands. Thresholds route work; they do not replace review.

B. Operational Metrics

Days of cover for part p is **DOC = available inventory / protected daily demand**. Available inventory excludes quarantined stock and can apply yield, expiry, and allocation factors. Projected shortage date is the first period where cumulative qualified supply falls below cumulative protected demand. Expected incremental cost for an action is **EIC = premium purchase + logistics + qualification + changeover + penalties - avoided loss**. Values are displayed as ranges where costs or recovery times are uncertain. Financial assumptions are stored separately from source facts.

C. Production Stoppage Probability

Production stoppage probability is estimated by a probabilistic simulation service using Monte Carlo sampling over distributions for disruption duration, recovery time, transit delay, yield, demand, and alternative capacity. The Analyst agent supplies validated scenario parameters, invokes the simulation service, and explains the results; the LLM does not perform the numerical simulation itself. The displayed value is the fraction of runs in which a protected line cannot meet its minimum schedule. Calibration is assessed on historical or synthetic labeled incidents; absent such evidence, the interface must label the output a scenario frequency rather than an empirical probability.

D. Response Generation

The Planner receives only verified facts, twin state, scenario results, and constraints. It generates a diverse set of candidate mitigation options, and invokes a deterministic checker which ranks the options by a multi-objective score:

$$U(a) = \beta_S \cdot \text{Service}(a) + \beta_F \cdot \text{NetFinancialImpact}(a) + \beta_R \cdot \text{ReducedRisk}(a) - \beta_I \cdot \text{ImplementationCost}(a)$$

The interface shows each component instead of hiding the weighting. A user can select service-first, cost-first, or balanced policy profiles. No alternative supplier is recommended solely because it appears in public text: the supplier must satisfy enterprise approval, technical qualification, sanctions/export-control checks, capacity, commercial, and cybersecurity requirements.

Action class	Feasibility checks	Approval
Expedite/reroute	lane, carrier, customs, cost	Logistics owner
Reallocate inventory	priority, contract, service	Operations
Switch supplier	quality, capacity, legal, IP	Cross-functional
Reschedule	line, tooling, labor, due dates	Plant planner
Customer notice	facts, disclosure, commitment	Account owner

TABLE VI. ACTION CONTROLS

E. RAG-Grounded Mitigation Planning

During mitigation planning, RAG retrieves authorized, current records—including approved suppliers, qualified substitute components, contracts, company policies, available capacities, logistics routes, and previous disruption responses—and provides them as evidence to the LLM-based Planner [9]. The LLM uses this retrieved context to reason about and propose candidate mitigation actions, while deterministic feasibility checks and optimization services validate constraints and compare the candidate options based on service recovery, implementation cost, net financial impacts, and reduced risk. Each recommendation retains links to the retrieved records so decision-makers can verify why the action was proposed; missing, stale, or conflicting records trigger review rather than unsupported recommendations.

F. Business Value

The value hypothesis has four components: reduced time-to-awareness; reduced time-to-impact assessment; lower expected disruption cost through earlier optionality; and more consistent, auditable communication. Benefits should be measured against a baseline workflow rather than inferred from model novelty.

A participating company provides authorized supply-chain data, operational constraints, and decision rules. Chainsilience AI converts these inputs into earlier warnings, quantified exposure, and feasible mitigation options, while the company retains decision authority and supplies outcome data that improves future scenario calibration. Value is measured through avoided expected disruption loss, analyst hours saved, and improved service continuity relative to implementation and operating cost.

For an SME, a useful deployment can begin with one product family, a limited supplier set, inventory and purchase-order extracts, and a small number of trusted external sources. This minimizes integration cost and gives the team a measurable pilot. Semiconductor deployments can then extend to multi-tier material, utility, equipment, foundry, assembly/test, and route dependencies.

G. Alert Content

A complete alert states: **signal** (what changed); **exposure** (the graph path); **impact** (orders, dates, cost, uncertainty); **options** (ranked and constrained); **decision** (owner and deadline); and **evidence** (sources and retrieved passages). The same structured object powers the dashboard, internal notification, and email draft, preventing channel-to-channel inconsistency.

Where sources disagree, the alert shows the conflict and requests confirmation. Where internal data are stale, it displays the last refresh time. Where no feasible mitigation exists, it recommends escalation rather than fabricating an alternative.

V. CASE STUDY AND EVALUATION

A. Evidence-Based Trigger

On 3 April 2024, a magnitude 7.4 earthquake occurred near Hualien City, Taiwan, followed by strong aftershocks [11]. TSMC later reported that earthquakes in April 2024 caused damage to inventories, facilities,

machinery, and equipment and that the company recognized approximately NT\$3 billion in losses, net of insurance claims, in the second quarter of 2024 [12]. These facts establish a real disruption context, but they do not establish the hypothetical buyer impacts below.

B. Illustrative Enterprise Scenario

Assume a fictional electronics SME, *Orion Controls*, buys a controller chip through Supplier A located in Taiwan. The enterprise twin states that Supplier A represents 42% of qualified wafer volume; usable inventory is 12 days; protected demand is 8,000 units/day; Supplier B is approved for 25% incremental volume after a five-day changeover; and premium logistics plus changeover would cost USD 180,000. These values are design-test inputs, not claims about any real company.

Stage	Illustrative output
Verify	M7.4 event; location/time confirmed; confidence 0.94
Map (Knowledge Graph)	Event -> Taiwan site -> Supplier A -> chips -> 3 SKUs -> 7 orders
Simulate	Baseline shortage day 13; 3-week recovery range
Estimate	71% chance of production stoppage within 21 days
Plan	Shift 25% to Supplier B; expedite 6 days; reallocate stock
Cost	USD 180k estimate; range USD 150k-230k

TABLE VII. ILLUSTRATIVE EVENT-TO-ACTION TRACE

The system would display the evidence supporting the earthquake, the enterprise records supporting the dependency and inventory figures, the simulation version, and the scenario frequency by Monte Carlo simulation producing the 71% chance of production stoppage. A user can compare 'wait,' 'expedite,' 'reallocate,' and 'switch 25%' alternatives before approving supplier outreach.

C. Example Communication

The Communicator drafts a supplier-status request that cites the affected part and asks for facility status, available capacity, recovery process, allocation policy, and next update time. It does not state that a three-week delay is confirmed. The procurement owner edits and approves the draft; the system records the sent version and any reply as new evidence.

D. Evaluation Design

Evaluation uses a replay corpus of dated disruption events and frozen enterprise snapshots. Articles arriving after each decision cutoff are withheld to prevent future information leakage. Two analysts independently label event fields and relevant graph paths; disagreements are adjudicated. Synthetic events supplement rare categories but are reported separately.

Dimension	Metric	Pilot target*
Detection	Detecting disruptions 24h; false alerts/day	>=85%; <=2
Extraction	Correct extraction of Entity/relation rate	>=85%
Verification	Unsupported-claim rate	<2%
Mapping	Disruption connected to correct supply chain path rate	>=80%
Risk ranking	Precision@K; rank correlation	Improve baseline
Production stoppage probability	Brier score; calibration error	Improve baseline
Impact	Accuracy for delay/DOC	Use-case set
Action	Feasible recommendation rate	>=90%
Workflow	Median alert-to-decision time	-50% vs baseline
Safety	Unauthorized action rate	0

TABLE VIII. PROPOSED PILOT METRICS. *Targets are acceptance criteria, not achieved results.

Baselines are: keyword alerts plus manual analysis; LLM extraction without verification; and risk ranking without a digital twin. Component-removal tests remove provenance, graph context, or deterministic simulation to quantify their contribution. Analysts blind-score relevance, traceability, feasibility, and clarity. Operational users complete timed scenario tasks and a structured usefulness survey.

E. Feasibility and Implementation

A 12- to 16-week pilot is proposed. Weeks 1-3 define the ontology, data-sharing boundary, and success metrics. Weeks 4-7 integrate a narrow supplier/product scope and build event extraction. Weeks 8-10 add graph mapping and deterministic inventory/lead-time calculations. Weeks 11-13 add scenario comparison, dashboard, and draft communications. Weeks 14-16 run replay tests, red-team exercises, and user acceptance.

The technical stack can use a relational store for enterprise records, a graph SQL database for dependencies, object storage for source documents, a queue for event processing, a simulation service, and a web dashboard. Model choice remains replaceable. Deployment may be cloud (via AWS), virtual private cloud, or on-premises depending on confidentiality and integration constraints.

F. Evaluation Threats

Historical coverage may favor highly reported events; ground truth for supplier impact may be unavailable; analysts may disagree on relevance; and synthetic scenarios may not match real behavior. Results therefore require confidence intervals, category-level reporting, dataset documentation, and clear separation between retrospective accuracy and prospective operational value.

VI. GOVERNANCE, LIMITATIONS, AND CONCLUSION

A. Data, Security, and Privacy

The governance design follows the NIST AI RMF functions - Govern, Map, Measure, and Manage [13] - and addresses generative-AI risks such as confabulation and information integrity [14]. Cybersecurity controls map to NIST CSF 2.0 [15]. Zero-trust principles require explicit, continuously evaluated access to resources rather than trust based on network location [16].

Controls include source allowlists; data minimization; encryption; tenant isolation; role- and attribute-based access; secret management; software and model inventories; prompt/input sanitization; network egress restrictions; immutable audit events; retention schedules; backups; incident response; and periodic penetration and dependency testing. Sensitive supplier prices, forecasts, and customer orders are excluded from prompts unless strictly required and contractually permitted.

B. Hallucination and Source Controls

Generated facts must cite retrieved evidence. Structured fields are schema-validated; numeric values are recalculated outside the LLM; dates and units are normalized; contradictions trigger review; and low-confidence mappings cannot automatically propagate to high-severity actions. The interface distinguishes quoted source facts, enterprise facts, model inferences, scenario assumptions, and user decisions.

C. Human Oversight and Ethics

Consequential decisions remain accountable to named roles. Procurement commitments, supplier changes, customer promises, and external emails require approval. Recommendations should be checked for systematic disadvantage to small or geographically concentrated suppliers, and users must not treat public reporting as proof of wrongdoing. Source licensing, privacy, export-control, sanctions, competition, and contractual duties require jurisdiction-specific review. Chainsillence AI must securely protect sensitive company data, including suppliers, routes, products, orders, inventory, and production schedules, using encryption and strict access controls. This information

must not be disclosed to third parties or used to train external AI models without explicit company consent.

D. Limitations

Chainsilience AI cannot observe undisclosed multi-tier dependencies, guarantee source truth, predict unprecedented events with precision, or make unqualified suppliers immediately usable. Digital twins decay when master data are stale. LLM outputs remain probabilistic; simulation outputs inherit uncertain distributions; and optimizing one firm's resilience can shift cost or scarcity to others. The system is decision support, not autonomous authority.

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E. Future Work

Future work includes using MCPs to connect LLM to email sending services such that an agent can autonomously send customer/supplier email after human approval, probabilistic entity resolution, richer multi-tier mapping, causal recovery models, federated or privacy-preserving learning, active learning from analyst corrections, robust optimization, multilingual evaluation, and standardized connectors. A mature deployment should measure not only detection quality but whether decisions reduce expected loss without creating unacceptable security, fairness, or compliance risk.

F. Conclusion

Chainsilience AI proposes a practical bridge between disruption news and operational response. Its architecture combines verified event extraction, provenance, a supply-chain knowledge graph, a digital twin, deterministic and probabilistic analytics, constrained planning, and approval-gated agentic communication. The design is deliberately hybrid: LLMs interpret language and draft explanations; graphs preserve relationships; simulators estimate consequences; solvers test feasible mitigations; and humans retain authority.

The central test for IDSOL 2026 is therefore not whether the system can produce a persuasive alert. It is whether that alert is earlier than the baseline, correctly linked to the company, quantitatively reproducible, operationally useful, secure, and safe to act upon. The proposed replay study and staged pilot make those claims falsifiable.

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